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Detecting Outliers

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1. A common data analysis task is outlier detection. One example would be if we are managing security on internet networks and would like to detect anomalies in the volume of data sent over our network nodes. Higher than usual traffic should be detected and flagged so that it can be investigated to ensure it is not being caused by malicious activities. **1 / 1 point**

   The standard deviation distance function created in the previous assignment can be used for such outlier detection tasks. It is common to label data as outlier if it lies more than a specified number of standard deviations from the mean, i.e., if the standard deviation distance to the mean is larger than a specified number. Typically this is chosen to be 3 or larger. Smaller numbers represent more typical deviations from the mean. Afterall, 1 standard deviation is intended to represent the most typical deviation from the mean, and therefore should not be considered an outlier.

   In the attached '.mat' file, you will find data vector **nodetraffic**, which represents recordings of the number of data packets per second travelling through a node.

   What is the mean of the dataset, rounded to the nearest integer?

   nodetraffic
   MAT File

   - ○ 14009
   - ◉ 9829
   - ○ 12657

   Correct

2. What is the standard deviation of the dataset, rounded to the nearest integer? **1 / 1 point**

   - ○ 12657
   - ○ 9829
   - ◉ 14009

   Correct

3. If we decide to classify the traffic for a node as suspicious if the traffic through the node has a standard deviation distance (see previous matlab grader problem on "standard deviation distance") to the mean that is larger than 6, how many instances in **nodetraffic** are suspicious and warrant further investigation? **1 / 1 point**

   24

   Correct